

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 7: High-Lift Devices (Flaps & Slats) & Contamination

Syllabus Source: CAE Oxford (pp. 225-264) | Pass Target: $\geq 90\%$ AviationExam

1. TRAILING-EDGE FLAPS: CAMBER & AREA EXPANSION

Trailing-edge flaps increase maximum lift coefficient (C_{L_max}) to reduce take-off and landing speeds:

Flap Type	Mechanism of Operation & Boundary Layer Control	Aerodynamic Characteristics
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Plain Flap	Simple hinged trailing edge deflects downward, increasing camber.	Increases C_{L_max} moderately; moderate drag increase.
Split Flap	Deflects lower surface downward while upper surface remains fixed.	Large suction wake behind split plate; massive parasite drag increase; ideal dive brake.
Slotted Flap	Leaves an aerodynamic converging slot between wing and flap.	High-pressure lower air shoots through slot as high-velocity jet, energizing flap boundary layer and delaying flow separation.
Fowler Flap	Translates REARWARD on tracks before deflecting downward.	INCREASES BOTH CAMBER AND WING SURFACE AREA (S)! Achieves the highest C_{L_max} increase of any flap design! Modern airliners use multi-slotted Fowler flaps.

- Aerodynamic Effects of Flap Extension: 1. Increases C_{L_max} (lowers stall speed V_S); 2. Increases Drag (steepens approach descent angle); 3. Generates NOSE-DOWN pitching moment (due to rearward movement of center of pressure); 4. REDUCES the critical stalling angle of attack (α_{crit} decreases by $\sim 2^\circ$ to 4° !).

2. LEADING-EDGE HIGH-LIFT DEVICES: SLATS & SLOTS

Leading-edge slats allow an aircraft to operate at dramatically higher angles of attack without stalling:

- Slats & Slots Principle: A leading-edge slat opens a convergent nozzle slot that ducts high-energy air onto the upper surface. This delays flow separation at high angles of attack.
- α_{crit} Extension: Unlike trailing-edge flaps (which decrease α_{crit}), LEADING-EDGE SLATS EXTEND THE STALLING ANGLE OF ATTACK from $\sim 15^\circ$ up to 25° or even 30° ! Slats do not change zero-lift AOA.
- Krueger Flaps: Folding leading-edge flaps that rotate out from beneath the lower wing leading edge. Used near the engine nacelles on Boeing 747/777/737.

3. AIRFRAME CONTAMINATION: THE CLEAN AIRCRAFT CONCEPT

Frost, ice, and snow clinging to aerodynamic surfaces degrade flight safety catastrophically:

- Surface Roughness Effect: Even a thin frost layer with the roughness of medium sandpaper:
 1. Reduces maximum lift (C_{L_max}) by 25% to 30%!
 2. Increases stall speed by 15% to 20%!
 3. Increases drag by over 100%!
 4. Drastically reduces stalling angle of attack.

- Clean Aircraft Concept (EASA Mandate): Take-off is STRICTLY PROHIBITED unless all critical aerodynamic lifting surfaces, control surfaces, and engine intakes are completely free from frost, ice, snow, and slush contamination!

[!] EXAM TRAP (AVIATIONEXAM) - Flaps vs Slats Stalling Angle of Attack Traps

- Trailing-Edge Flaps: INCREASE C_{L_max} , but DECREASE the stalling angle of attack (α_{crit} drops by 2-4°)! - Leading-Edge Slats: INCREASE C_{L_max} , and DRAMATICALLY INCREASE the stalling angle of attack (α_{crit} increases from ~15° to 25°+)! - Frost / Ice on Wings: Reduces C_{L_max} AND reduces α_{crit} (wing can stall prematurely at take-off pitch attitude)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Slats vs Flaps Operational Distinction

Exam Query: Which high-lift device increases C_{L_max} while reducing the stall angle of attack? Step 1: Check effect on critical angle. If it REDUCES critical angle -> It is a trailing edge FLAP. Step 2: If it INCREASES critical angle (delays stall to higher angles up to 25 degrees) -> It is a leading edge SLAT. Memory Rule: Slats = Super Angle (increases stall angle). Flaps = Fast Lift (reduces stall speed and stall angle).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What happens to the critical stalling angle of attack (α_{crit}) when TRAILING-EDGE FLAPS are extended?

- [A]: Stalling angle of attack increases
- [B]: Stalling angle of attack DECREASES (wing stalls at a lower AOA than clean)
- [C]: Stalling angle remains unchanged
- [D]: Stalling angle becomes 45°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Extending trailing-edge flaps increases camber, raising the upper surface adverse pressure gradient. This causes the boundary layer to separate earlier, reducing the critical stalling AOA by 2° to 4°.

Q2: What is the primary aerodynamic mechanism of a Fowler flap compared to a plain flap?

- [A]: It reduces parasite drag
- [B]: It extends rearward on tracks before deflecting, increasing both wing surface area (S) and camber, providing superior C_{L_max} increase
- [C]: It prevents Dutch roll
- [D]: It trims the rudder

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Fowler flaps translate aft on tracks to increase effective wing planform area before deflecting down to increase camber, yielding the highest C_{L_max} increase of any flap type.

Q3: How does a thin layer of frost on the wing upper surface affect take-off aerodynamic characteristics?

- [A]: It has zero effect

- [B]: It reduces maximum lift by up to 30%, increases stall speed significantly, and reduces the critical stalling angle of attack
- [C]: It lowers fuel consumption
- [D]: It acts as a laminar flow generator

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Frost acts like coarse sandpaper, tripping the laminar boundary layer prematurely into a thick, low-energy turbulent layer that separates prematurely, causing stall at normal take-off rotation angles.